WORKSHEET 2.7

Differentiating Products and Quotients

Unit 2: The Derivative

Key ideas

- **Product rule**: $(fg)' = f'g + fg'$. The derivative of a product is *not* the product of the derivatives.
- **Quotient rule**: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$. The order in the numerator matters.
- With a table of values, substitute the numbers straight into the rule.
- If a product or quotient can be simplified by algebra first, that is usually quicker.

Worked examples

Example 1 · A product

Find $\frac{d}{dx}\left[x^3 e^x\right]$.

Solution. $3x^2e^x + x^3e^x$.

$e^x\left(x^3 + 3x^2\right)$.

Example 2 · A quotient

Find $\frac{d}{dx}\left[\frac{2x+1}{x-3}\right]$.

Solution. $\frac{2(x-3) - (2x+1)(1)}{(x-3)^2}$.

$-\frac{7}{(x-3)^2}$.

Example 3 · From values

Given $f(1) = 2$, $f'(1) = 3$, $g(1) = -4$ and $g'(1) = 5$, find $(fg)'(1)$ and $\left(\frac{f}{g}\right)'(1)$.

Solution. $(fg)'(1) = 3(-4) + 2(5)$. $\left(\frac{f}{g}\right)'(1) = \frac{3(-4) - 2(5)}{(-4)^2}$.

-2 **and** $-\frac{11}{8}$.

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Practice

CORE

1. Find $\frac{d}{dx}[x \sin x]$.

2. Find $\frac{d}{dx}[x^2 \ln x]$.

3. Find $\frac{d}{dx}\left[\frac{x}{x+1}\right]$.

4. Find $\frac{d}{dx}[e^x \cos x]$.

EXAM LEVEL

5. Find $\frac{d}{dx}\left[\frac{\cos x}{x}\right]$.

6. Find $\frac{d}{dx}\left[\frac{x^2-1}{x^2+1}\right]$ in simplified form.

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7. A differentiable function f has $f(2) = 8$ and $f'(2) = -4$. Let $h(x) = \dfrac{f(x)}{x^2}$. Find $h'(2)$.

8. At what value of x does the graph of $y = (x-3)e^x$ have a horizontal tangent?

Question 9 · Free-response practice

Let $f(x) = (x^2 - 3)\, e^x$.

(a) Find $f'(x)$ in factored form.

(b) Find all values of x at which the graph of f has a horizontal tangent.

(c) Write an equation of the line tangent to the graph of f at $x = 0$.

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Worked solutions

1. $\sin x + x\cos x$.
2. $2x\ln x + x$.
3. $\dfrac{(x+1)-x}{(x+1)^2} = \dfrac{1}{(x+1)^2}$.
4. $e^x\cos x - e^x\sin x = e^x(\cos x - \sin x)$.
5. $\dfrac{-x\sin x - \cos x}{x^2}$.
6. $\dfrac{2x(x^2+1)-2x(x^2-1)}{(x^2+1)^2} = \dfrac{4x}{(x^2+1)^2}$.
7. $h'(x) = \dfrac{x^2f'(x)-2xf(x)}{x^4}$, so $h'(2) = \dfrac{4(-4)-4(8)}{16} = -3$.
8. $y' = e^x + (x-3)e^x = (x-2)e^x$, which is zero at $x = 2$.
9. (a) $2xe^x + \left(x^2-3\right)e^x = \left(x^2+2x-3\right)e^x = (x+3)(x-1)e^x$.
   (b) $x = -3$ and $x = 1$, since e^x is never zero.
   (c) $f(0) = -3$ and $f'(0) = -3$: $y = -3x - 3$.

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